Ex 5

E-field of a conducting sphere:

$$E = \frac{Q}{4\pi\epsilon_0 r^2} \Rightarrow Q = 4\pi\epsilon_0 r^2 E$$

at the surface:

$r_0 = 6371$ km, $E_0 = -130$ V/m

$$\Rightarrow Q = 4\pi\epsilon_0 r_0^2 E_0^2 = -5.87 \times 10^5 \text{ C}$$

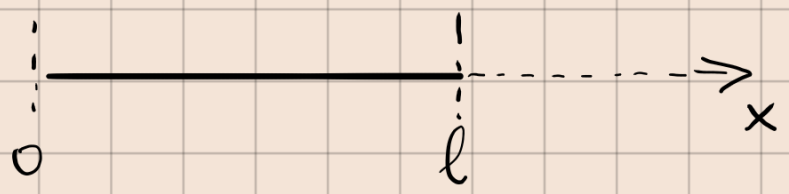
The charge is negative because we know that the field points to the center.

Ex 6

a) λ: Charge per length

total charge: $q = \lambda \cdot l = 17.5\ nC$

b)



$l = 5m$

E-field at position x:

$$E(x) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{x(x-l)}$$

at $x = 6m$: $E(6m) = 26\ N/C$

at $x = 9m$: $E(9m) = 4.4\ N/C$

Ex 7

a) Potential of a point charge:

$$U(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r}$$

for a proton:

$$U_p(r) = \frac{1}{4\pi\epsilon_0} \cdot \frac{e}{r} = 14.38\ V$$

b) Potential energy $W = q \cdot U$

For the electron-proton system:

$$W = q_e \cdot U_p(r) = (-1e) \cdot (14.38\ V)$$

$$= 14.38\ eV$$

Conversion: $1eV = (1e) \cdot V$

$$= (1.6 \times 10^{-19}\ C) \cdot V$$

$$= 1.6 \times 10^{-19}\ J$$

$\Rightarrow \quad W = -14.38 \text{ eV} = -2.3 \times 10^{-18} \text{ J}$

c) We just change r to 5.3×10^{-11} m.

$\Rightarrow$ Potential: $\quad U_p(r) = 27.13 \text{ V}$

$\Rightarrow$ Energy: $\quad W = q_e \cdot U_p(r) = -27.13 \text{ eV}$

$= -4.3 \times 10^{-18} \text{ J}$

Ex 8

a) Electron accelerated in E-field:

Force: $F = qE$

Newton: $F = ma$

$\Rightarrow qE = ma$

$\Rightarrow E = \frac{ma}{q} = 11.3 \times 10^3 \ V/m$

$= 11.3 \ kV/m$

b) Remember velocity from last semester:

$v = at \Rightarrow t = \frac{v}{a} = 10^{-9} \ s$

$= 1 \ ns$.